WORKED SOLUTION

STEP I 1989, Question 13

Mechanics • Ladder Equilibrium with Friction

STEP I 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

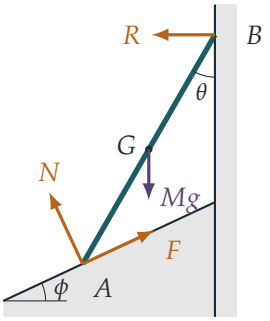
step.maths.org/questions/89-s1-q13

The key idea

This is a three-force-plus-friction equilibrium problem, and the trick is to choose good directions. Taking moments about the foot of the ladder removes both N and F and gives the wall reaction at once. Then, instead of resolving horizontally and vertically, we resolve *perpendicular to and along the slope*. Each of those equations contains only one of the unknowns N and F, so the given formulas drop straight out.

Solution

We model the ladder as a uniform rod AB of length $2a$ with its centre of mass G at the midpoint. The foot A is on the rough slope, and the top B is against the smooth wall, so the wall's reaction R is horizontal, pointing away from the wall. At A there is a normal reaction N perpendicular to the slope and a friction force F along it. The ladder would tend to slip with its foot sliding down the slope, away from the wall, so we take F as acting up the slope towards the wall. (The formula for F turns out positive, which confirms the choice.)



The wall reaction

Taking moments about A removes N and F. The ladder makes angle θ with the vertical, so B is a height $2a\cos\theta$ above A and G is a horizontal distance $a\sin\theta$ from A. The horizontal force R at B has moment $R \cdot 2a\cos\theta$, and the weight has moment $Mg \cdot a\sin\theta$ in the opposite sense, so

$$2aR\cos\theta = Mga\sin\theta \qquad \Longrightarrow \qquad R = \tfrac{1}{2}Mg\tan\theta.$$

Resolving perpendicular to and along the slope

> **Ask yourself** Which directions should we resolve in, so that each equation contains only one unknown?

Resolve perpendicular to the slope and along it. F has no component perpendicular to the slope, and N has no component along it. We need the components of the other two forces.

- The weight Mg has component $Mg\cos\phi$ into the slope and $Mg\sin\phi$ down the slope.
- The slope rises towards the wall, so its upward normal leans away from the wall, and the horizontal force R (also pointing away from the wall) has component $R\sin\phi$ along that normal, i.e. *out* of the slope. Along the slope, R points away from the wall, which is downhill, with component $R\cos\phi$.

Perpendicular to the slope:

$$N + R\sin\phi = Mg\cos\phi,$$
$$\text{so} \quad N = Mg\cos\phi - \tfrac{1}{2}Mg\tan\theta\sin\phi = Mg\left(\cos\phi - \frac{\tan\theta\sin\phi}{2}\right).$$

Along the slope (up the slope positive):

$$F = Mg\sin\phi + R\cos\phi \quad \Longrightarrow \quad F = Mg\left(\sin\phi + \frac{\tan\theta\cos\phi}{2}\right).$$

These are the two required results.

The largest angle

For the foot not to slip we need the contact to hold, $N > 0$, and friction to be limited: $F \le \mu N$ with $\mu = 2$. (Here $F > 0$, so this is the only friction condition.) With $\phi = 45^\circ$ we have $\sin\phi = \cos\phi = \frac{1}{\sqrt{2}}$, and writing $t = \tan\theta$,

$$F = \frac{Mg}{\sqrt{2}}\left(1 + \frac{t}{2}\right), \qquad N = \frac{Mg}{\sqrt{2}}\left(1 - \frac{t}{2}\right).$$

So $N > 0$ requires $t < 2$, and then $F \le 2N$ becomes

$$1 + \frac{t}{2} \le 2 - t \quad \Longleftrightarrow \quad \frac{3t}{2} \le 1 \quad \Longleftrightarrow \quad \tan\theta \le \frac{2}{3}.$$

This is stricter than $t < 2$, so it is the binding condition. Since $\tan\theta$ is increasing on $0 < \theta < 90^\circ$, the largest angle for which the ladder can rest in equilibrium is

$$\theta = \arctan\tfrac{2}{3} \approx 33.7^\circ,$$

and at that angle friction is limiting.

Checks

- With $\phi = 0$ (level ground) the formulas give $N = Mg$ and $F = \frac{1}{2}Mg\tan\theta = R$, the familiar results for a ladder on a horizontal floor.
- With $\theta = 0$ (ladder vertical) the wall does nothing, and $N = Mg\cos\phi$, $F = Mg\sin\phi$, just as for a particle resting on the slope.
- Numerically, solving the horizontal and vertical force equations directly for random θ and ϕ agrees with both formulas to 10^{-13}, and with $\phi = 45^\circ$ the ratio F/N passes through 2 at $\theta = 33.690^\circ$.

Common slips

- Getting the sign of the $R\sin\phi$ term wrong. Because the slope rises towards the wall, R partly pulls the foot *off* the slope, which reduces N.
- Resolving horizontally and vertically, then making errors solving the resulting simultaneous equations. It works, but resolving along and perpendicular to the slope is much quicker.

Beyond STEP

Notice that the steeper the slope, the less the ladder can lean, because the wall reaction reduces N while adding to F. Problems like this are the start of statics, where we choose axes and moment points to decouple the unknowns. The same idea underlies the analysis of trusses and frameworks, and of stability more generally. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body equilibrium and friction carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.